INCH-POUND

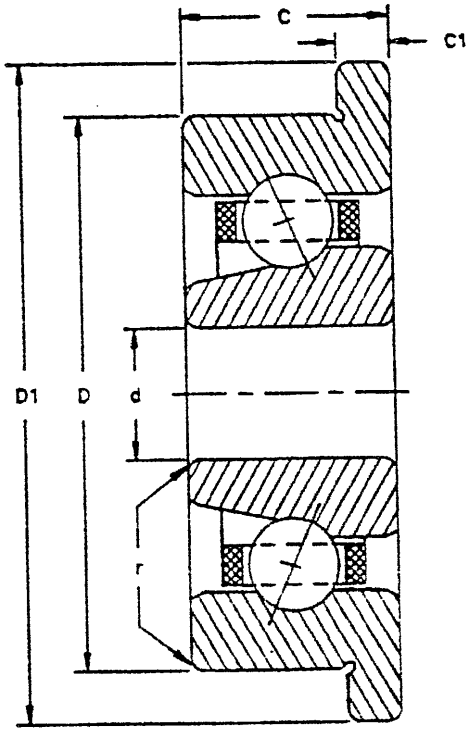
MIL-B-81793/29
15 March 1993

MILITARY SPECIFICATION SHEET

BEARING, BALL, ANNULAR, FOR INSTRUMENTS AND PRECISION ROTATING COMPONENTS, ANGULAR CONTACT, FLANGED, SEPARABLE AND STEPPED INNER RING, CHROMIUM-ALLOY STEEL, ABEC 5P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specification and Standards (DODISS) specified in the solicitation: MIL-B-81793



BEARINGS CONFIGURATIONS

AMSC N/A

FSC 3110

DISTRIUBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:11Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

1. Design. All bearings described by this specification sheet shall be angular contact instrument bearings with an unflanged, separable and stepped inner ring configuration.

2. Material. The ball and ring material for these bearings shall be chromium-alloy steel 52100 (UNS G52986) conforming to AMS 6444.

3. Tolerance class. The tolerance class shall be in accordance with the ABEC 5P tolerance tables of ANSI/AFBMA Standard 12.2. This tolerance class shall apply to all bearing sizes listed in Table I of this specification sheet.

4. Boundary dimensions. The boundary dimensions shall be as specified by the dash number (DASH NO) in Table I.

5. Retainer. The retainer shall be as specified by the part number designator (PN DES) in Table II.

6. closures. Closures are not available for this type of bearing. Part number designator N shall be used as shown in Table III.

7. Radial internal clearance or contact angle. Radial internal clearance or contact angle shall be as specified by the part number designator in Table IV.

8. Calibration. Calibration shall be as specified by the part number designator in Table V.

9. Performance Test. The performance test shall be as specified by the part number designator in Table VI.

10. lubrication.

10.1. Lubricant. The lubricant shall be in accordance with the specification indicated by the part number designator in table VII.

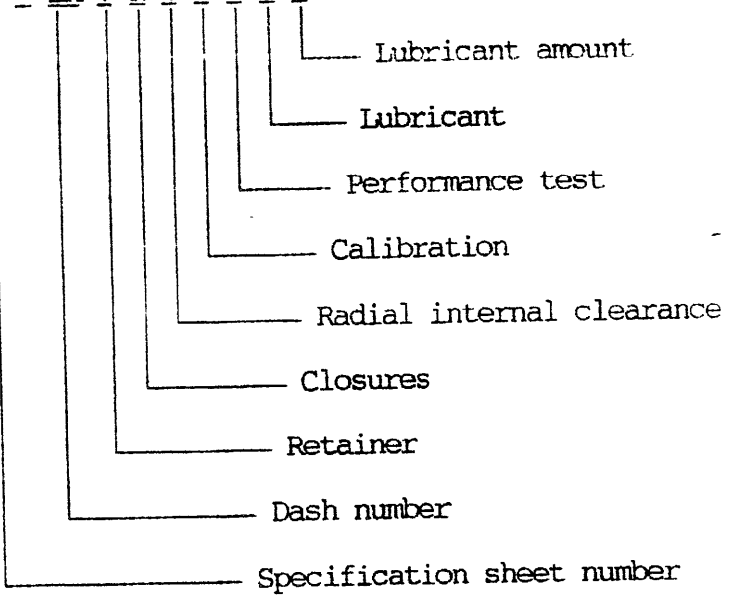
10.2. Lubricant amount. The amount of lubricant shall be as specified by the part number designator in Table VIII.

10.3. Barrier coating. Barrier mating shall be applied to bearings as specified by the part number designator in Table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:11Z
Check the source to verify that this is the current version before use.

11. Part number. The part number consists of the following: prefix letter M; general specification number; number of this specification sheet; dash number; and characters for retainer, closures, radial internal clearance or contact angle, calibration, performance test, lubricant, and lubricant amount.

EXAMPLE: M 81793/ 29 - BA 7 N 4 N 2 F 1



M81793/29-M7N4N2F1 indicates - Bore 0.0469″; outside diameter 0.1562″; width 0.0625″; flange OD 0.203″; flange width 0.013″; radius 0.003″; retainer, phenolic laminate; Closures, none; contact angle, greater than 14° but less than 20°; no calibration; performance test, starting torque - Grade A; lubricant, MIL-L-81322; lubricant amount, manufacturer's standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:11Z
Check the source to verify that this is the current version before use.

TABLE I. Boundary dimensions (inches).

DASH NO	BORE	OD	WIDTH	FLANGE OD	FLANGE WIDTH	RADIUS
	d	D	C	D_1	C_1	r 1/ 2/
-AA	0.0400	0.1250	0.0469	0.171	0.013	0.003
-BA	0.0469	0.1562	0.0625	0.203	0.013	0.003
-BB	0.0469	0.1562	0.0937	0.203	0.031	0.003
-CA	0.0550	0.1875	0.0781	0.234	0.031	0.003
-CB	0.0550	0.1875	0.1094	0.234	0.031	0.003
-DA	0.0781	0.2500	0.0937	0.296	0.023	0.003
-DB	0.0781	0.2500	0.1406	0.296	0.031	0.003
-EA	0.0937	0.1875	0.0625	0.234	0.018	0.003
-EB	0.0937	0.1875	0.0937	0.234	0.031	0.003
-EC	0.0937	0.3125	0.1094	0.359	0.023	0.003
-ED	0.0937	0.3125	0.1406	0.359	0.031	0.003
-FA	0.1250	0.2500	0.0937	0.296	0.023	0.003
-FB	0.1250	0.2500	0.1094	0.296	0.031	0.003
-FC	0.1250	0.3125	0.1094	0.359	0.023	0.003
-FD	0.1250	0.3125	0.1406	0.359	0.031	0.003
-FE	0.1250	0.3750	0.1094	0.422	0.023	0.003
-FF	0.1250	0.3750	0.1406	0.422	0.031	0.033
-FG	0.1250	0.3750	0.1562	0.440	0.030	0.012
-GA	0.1562	0.3125	0.1094	0.359	0.023	0.003
-GB	0.1562	0.3125	0.1250	0.359	0.036	0.003
-GC	0.1562	0.3125	0.1094	0.359	0.023	0.003
-GD	0.1562	0.3125	0.1250	0.359	0.036	0.003
-HA	0.1875	0.3750	0.1250	0.422	0.023	0.003
-HB	0.1875	0.3750	0.1250	0.422	0.031	0.003
-HC	0.1875	0.5000	0.1562	0.565	0.042	0.012
-HD	0.1875	0.5000	0.1960	0.565	0.042	0.012
-JA	0.2500	0.3750	0.1250	0.422	0.023	0.003
-JB	0.2500	0.3750	0.1250	0.422	0.036	0.003
-JC	0.2500	0.5000	0.1250	0.547	0.023	0.005
-JD	0.2500	0.5000	0.1875	0.547	0.045	0.005
-JE	0.2500	0.6250	0.1960	0.690	0.042	0.012
-KA	0.3125	0.5000	0.1562	0.547	0.031	0.005
-LA	0.3750	0.8750	0.2188	0.969	0.062	0.016
-LB	0.3750	0.8750	0.2812	0.969	0.062	0.016
-MA	0.5000	1.1250	0.2500	1.225	0.062	0.016
-MB	0.5000	1.1250	0.3125	1.225	0.062	0.016

1/ Maximum shaft or housing fillet radius that bearing corners will clear.

2/ Narrow face of rings shall have a maximum "r" value of one half the value listed provided this quantity does not fall below 0.003 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:11Z
Check the source to verify that this is the current version before use.

TABLE II. Retainer.

PN DES	TYPE
6	PTFE TOROIDS 1/
7	PHENOLIC TYPE LAMINATE 2/, 4/
8	VACUUM IMPREGNATED PHENOLIC 3/, 4/
9	NON-POROUS, NON-METALLIC CROWN 4/

1/ PTFE (Polytetrafluoroethylene).
2/ Phenolic or other porous non-metallic material.
3/ Phenolic or other porous non-metallic material saturated with lubricant. Selection of this choice dictates choosing manufacturer's standard amount of lubricant.
4/ Used for high speed applications.

TABLE III. Closures.

PN DES	NUMBER	TYPE
N	NONE	NONE

TABLE IV. Radial internal clearance or contact angle.

PN DES	RANGES
1	0.0001" to 0.0003"
2	0.0002" to 0.0005"
3	14° OR LESS 1/
4	GREATER THAN 14° BUT LESS THAN 20° 1/
5	0.0005" to 0.0008"
6	GREATER THAN 20° 1/
8	0.0008" to 0.0011"

1/ Angular contact bearing designations only.

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:11Z
Check the source to verify that this is the current version before use.

TABLE V. Calibration of bore and outside diameter.

PN DES	BORE INCREMENTS	OD INCREMENTS
N	NO CALIBRATION	NO CALIBRATION
A	NO CALIBRATION	0.00010
B	NO CALIBRATION	0.00005
C	0.00010	NO CALIBRATION
D	0.00010	0.00010
E	0.00010	0.00005
F	0.00005	NO CALIBRATION
G	0.00005	0.00010
H	0.00005	0.00005

TABLE VI. Performance test.

PN DES	TYPE
1	MANUFACTURER'S STANDARD
2	STARTING TORQUE 1/

1/ Starting torque limits from table III of MIL-B-81793.

TABLE VII. Lubricant.

PN DES	SPECIFICATION
P	PRESERVATIVE 1/
A	MIL-L-6085
B	MIL-L-6085 2/
C	MIL-L-81846
D	MIL-L-81846 2/
E	MIL-G-23827
F	MIL-G-81322
G	MIL-G-81937
H	MIL-G-83261
J	MIL-S-81087
K	MIL-S-81087 2/

1/ PN DES "P" shall be used only with PN DES "P" in TABLE VIII.
2/ With barrier coat.

Source: https://assist.dla.mil -- Downloaded: 202610-23T06:11Z
Check the source to verify that this is the current version before use.

TABLE VIII. Lubricant amount.

PN DES	QUANTITY
P	PRESERVATIVE AMOUNT REQUIRED BY MIL-P-197
1	MANUFACTURER'S STANDARD 1/
2	OIL PER TABLE II of MIL-B-81793
3	15% GREASE PER TABLE II of MIL-B-81793
4	25% GREASE PER TABLE II of MIL-B-81793
5	35% GREASE PER TABLE II of MIl-B-81793
6	45% GREASE PER TABLE II of MIL-B-81793

1/ Grease, fill to minimum 25%, maximum 40% of bearing void. Oil, immerse and then allow excess to drip off. The standard quantity of oil varies with each bearing size.

Custodians:
Army-AT
Navy-As
Air Force - 99

Review activities:
Navy-SH
DLA- IS

User activities:
Navy-MC
Air Force - 84

Preparing activity:
Navy - AS

Agent:
DLS-IS

(Project 3110-0823-29)

Source: https://assist.dla.mil -- Downloaded: 2026l0-23T06:11Z
Check the source to verify that this is the current version before use.